

- iii. **Atoll** A reef of narrow growing corals of horse—shoe shape and crowned with palm trees is called an Atoll.
- Atoll partly or completely encloses a lagoon. It is formed around an island in an elliptical form on a submarine platform. Funafuti atoll of Ellice Islands is a famous atoll.

CORAL BLEACHING

When corals are stressed by changes in conditions such as *temperature*, *light* or *nutrients*, they expel the symbiotic algae living in their tissues, causing them to turn colourless, this phenomenon is known as Coral Bleaching.

SALINITY

Salinity is defined as the total amount of salts and minerals in grams contained in 1 kg of sea water and is expressed as part per thousand. The oceanic salinity not only affects the marine organism and plant community, but also affects the physical properties of the ocean such as temperature, pressure, density, waves and currents.

- Average salinity of seawater is about 35‰ and average salinity in Southern hemisphere is more than that of Northern hemisphere.
- **Iso-halines** represent the salinity distribution in the surface of the sea. These are the lines joining places having an equal degree of salinity.
- The main source of salinity is dissolution of the rocks of oceanic crust, which contains salts. Poles have minimum salinity because of addition of fresh water in the form of icebergs and excessive snowfall.
- Variation in salinity causes vertical circulation of water.
- Salinity also increases the density of water. More saline water freezes slowly while the boiling point of saline water is higher than the fresh water.

Controlling Factor	Relation with Salinity
Evaporation	Greater the evaporation, higher the salinity.
Precipitation	Higher the precipitation, lower the salinity.
Influx of river water	Big voluminous rivers pour down immense volume of fresh water into the oceans and salinity is reduced at the mouth.
Atmospheric pressure	Anticyclonic conditions with stable air and high temperature increases the salinity of the surface water of the ocean.
Circulation of oceanic water	Ocean currents affect the spatial distribution of salinity by mixing sea waters.

- Salinity on an average decreases from equator to poles. The highest salinity is recorded near the tropics rather than the equator because of the heavy precipitation in the equatorial region.

Straits

A strait is a narrow body of water that connects two larger bodies of water and separates two landmasses.

Name	Joins	Location
Malacca Strait	Andaman Sea and South China Sea	Indonesia-Malaysia
Palk Strait	Palk Bay and Bay of Bengal	India-Sri Lanka
Sunda Strait	Java Sea and Indian Ocean	Indonesia
Yucatan Strait	Gulf of Mexico and Caribbean Sea	Mexico-Cuba
Mesina Strait	Mediterranean Sea	Italy-Sicily
Bab-el-Mandeb Strait	Red Sea and Gulf of Aden	Yemen-Djibouti
Cook Strait	South Pacific Ocean	New Zealand (N and S Islands)
Mozambique Strait	Indian Ocean	Mozambique-Malagasy
Bass Strait	Tasman Sea and South Sea	Australia
Bering Strait	Bering Sea and Chukchi Sea	Alaska-Russia
Bosphorus Strait	Black Sea and Marmara Sea	Turkey
Davis Strait	Baffin Bay and Atlantic Ocean	Greenland-Canada
Denmark Strait	North Atlantic and Arctic Ocean	Greenland-Iceland
Dover Strait	English Channel and North Sea	England-France
Florida Strait	Gulf of Mexico and Atlantic Ocean	USA-Cuba
Hormuz Strait	Gulf of Persia and Gulf of Oman	Oman-Iran
Hudson Strait	Gulf of Hudson and Atlantic Ocean	Canada
Gibraltar Strait	Mediterranean Sea and Atlantic Ocean	Spain-Morocco
Magellan Strait	Pacific and South Atlantic Ocean	Chile
Tsugaru Strait	Japan Sea and Pacific Ocean	Japan (Hokkaido-Honshu Island)

MOVEMENTS OF OCEANIC WATER

Waves

- Waves are the oscillatory movements in water mainly produced by winds, manifested by an alternate rise and fall of the sea surface. The waves are the important agents of erosion in the coastal regions, where they carve out various landforms like caves, bays, gulfs, capes and cliffs.
- Seismic waves or tsunamis are the waves caused by earthquakes in volcanic eruptions in the sea bottom. The tsunamis, which hit the coast in South-East Asia on 26th December, 2004, caused havoc in that region.

Ocean Currents

- An ocean current is a continuous, directed movement of ocean water generated by the forces acting upon it, such as breaking waves, wind, coriolis effect, temperature and salinity differences shape and configuration of the coastline and tides caused by the gravitation pull of the Moon and the Sun.
- Ocean currents circulate in clockwise direction in Northern hemisphere and in anti-clockwise direction in Southern hemisphere. *Ocean currents are of two types :*
 - i. **Warm Ocean Currents** The ocean currents flowing from lower latitude to higher latitude are called as warm currents.
 - ii. **Cold Ocean Currents** The ocean currents flowing from higher latitude to lower latitude are called as cold currents.

Currents of North Pacific Ocean

(i) Warm Currents

- **North Equatorial Current** It flows Westwards from the Western coast of Mexico to the Philippines.
- **Kuroshio Current** It is an extension of North Equatorial current near Japan coast. It flows towards North.
- **Alaska Current** It flows along the coast of British Columbia and the Alaska Peninsula.

(ii) Cold Currents

- **Oyashio Current** It flows down from Bering sea towards Japan from North pole and it joins Kuroshio currents.
- **Okhotsk Current and Kuril Current** It flows down from sea of Okhotsk and joins Kuroshio current to the North of Japan.
- **West Wind Drift** It flows towards Alaska.
- **Californian Current** It is an extension of Alaskan currents. It finally joins North Equatorial current and completes clockwise circulation of water.

Currents of South Pacific Ocean

(i) Warm Currents

- **East Australian Current or Great Barrier Current** It flows towards East coast of Australia from equator towards Pole.
- **South Equatorial Current** It originates due to South East trade winds and flows Westwards and bifurcates near New Guinea.
- **Counter Equatorial Current** It extends upto Panama Bay. It flows exactly on equator from West to East.

(ii) Cold Currents

- **Peruvian Current** (Humboldt current) It flows from South towards equator on the coast of Chile and Peru.
- **West Wind Drift** It flows from Tasmania to Chile coast of South America.

Currents of North Atlantic Ocean

(i) Warm Currents

- **North Equatorial Current** It is present between equator and 10°N.
- **Cayenne Current** It flows adjacent to French Guinea and enters into Caribbean Sea and Gulf of Mexico.
- **Florida Current** Cayenne current near Florida (US coast) is called Florida Current.
- **Antilles Current** It flows to the East of West Indies and other islands.
- **Gulf Stream** It flows from US coast towards North-West Europe under the influence of Westerly winds.
- **North Atlantic Drift** Gulf Stream bifurcates into
 - North Atlantic drift (warm).
 - West Wind drift (cold) and Canaries current (cold).

(ii) Cold Currents

- **Labrador Current** It originates in Baffin bay and Davis Strait and merges with Gulf Stream near Newfoundland. Newfoundland is a famous zone of fishing, commonly known as Grand Bank.
- **Irminger Current or Greenland Current** It flows between Greenland and Iceland and merges with North Atlantic drift.
- **Canaries Current** It flows along the Western coast of North Africa between Madeira cape verde and it joins North equatorial current.
- **West Wind Drift** It flows towards UK.

Currents of South Atlantic Ocean

(i) Warm Currents

- **South Equatorial Current** It flows between equator and 10°S.
- **Brazilian Current** It flows to the East coast of Brazil from equator towards pole.

(ii) Cold Currents

- **Falkland Current** It flows along the South-East coast of South America from South to North.
- **Benguela Current** It flows from South to North near the 'Cape of Good Hope'.
- **West Wind Drift** It is the continuence of Brazilian and Falkland current.
- **Guinea Current** It flows near coast of Guinea (Africa).

Currents of the Indian Ocean

- **The Asiatic Monsoon** It influences the currents of the North Indian Ocean, while the currents of South Indian ocean are influenced by the atmosphere's anticyclonic circulation.

- **North Equatorial Current** The current flows from East to West and upon reaching the East coast of Africa, a good portion turns Southward, crosses the equator and becomes the Mozambique current.
- **Mozambique Current** The Mozambique current flows South along the East coast of Africa from the vicinity of the equator to about 35°, where it is known as Agulhas stream.
- **Agulhas Stream** The Agulhas stream flows Westward along the Southern West of Madagascar and joins the Mozambique current along the East African coast.
- **West Wind Drift Current** It flows across the Indian ocean to the South-West of Australia.
- **North-East Monsoon Drift** It flows along the coast of India during winter.
- **South-West Monsoon Drift** It flows along the coast of India during summer.
- **South Equatorial Current** It is a significant Pacific, Atlantic and Indian ocean current that flows East-West between the equator and about 20 degrees South.
- **Somalia Current** It is an ocean boundary current that runs along the coast of Somalia and Oman in the Western Indian ocean and is analogous to the Gulf stream in the Atlantic ocean.

Tides

The rises and fall of the sea level as a result of the forces between the Earth, the Moon and the Sun is called a Tide. The interval between two tides is 12 hrs and 26 min.

There are two types of tides, which are as follows :

- Spring Tides** When the Earth, the Moon and the Sun are in a straight line (also called SYZYG (straight line configuration of three celestial bodies) the Sun enhances the gravitational pull of the Moon, creating a condition of higher high tides and lower low tides known as Spring Tides.
- Neap Tides** When the Sun and Moon are at right angles to the Earth. The Sun partially contracts the pull of the Moon producing lower high tides typical of a Neap Tide.

Natural Vegetation

Natural vegetation refers to those vegetations which grows naturally on the Earth's surface without any human interference. Different vegetation regions that are identified have distinct types of plants, soil and weather patterns. *Major vegetation types of the world has been discussed below :*

Tropical Evergreen Rainforests

These vegetations are characterised by long and evergreen trees. Forest of these types have layered structure that form canopy at the top. Almost two-thirds of all the world's flowering plants are found in rainforests.

Hardwood trees like rosewood, teak, sal, ebony and mahogany are most common trees. Such forests are found near the equator in amazon basin, zaire basin and South-East Asian region.

Tropical Deciduous Forests

These forests are also known as Monsoon Forests. These forests are found in regions having monsoon climate i.e. India, Myanmar, Thailand, parts of Indonesia, Malaysia, Indo-China, Philippines etc. Trees are adapted to a long dry season and shed their leaves during dry season. Teak, sal, bamboo, khair, bargad, peepal are important trees. Forest are not very dense, therefore, commercial exploitation of these forests is possible.

Mediterranean Forests

These forests are found in areas around Mediterranean Sea, Central Chile, South-West USA, Australia and Northern Africa. Such vegetations are found in regions with hot and dry summers and mild and wet winters. Xerophytic plants constitute the natural vegetation. Mediterranean regions are known as 'Orchards of the World' for their fruit cultivation. Citrus fruits like oranges, figs, olives and grapes are commonly cultivated.

Coniferous Forests

These forest extend in a band across Northern North America and Eurasia to the edge of the Arctic Tundra, constituting the largest terrestrial biome in the world. Forests consist of tall, softwood evergreen trees. Northern coniferous forests are dominated by cone-bearing trees, such as pine, spruce, fir and hemlock. Trees are conical shaped with needle shaped leaves and trees are found in large groups.

Tropical Grasslands

These grasslands are known as 'Savanna' in Africa and 'Campos' in Brazil. The grasses are very tall and grows upto a height of 3 to 4 metres. The scattered trees found at different densities are thorny and have small leaves, an apparent adaptation to relatively dry conditions. Fires are common in dry season and dominant plant species are fire-adapted and tolerant of seasonal drought.

Temperate Grasslands

These grasslands are also known as Prairies in North America, Pampas in South America, Downs in Australia, Velds in South Africa and Steppes in Europe. These regions have warm and moist summers and cold and snowy winters. Dominant plants are grasses and forbs, which vary in height from few centimetres to 2 m. These areas are also known as 'Bread Basket' of the world due to large scale cultivation of maize and wheat.

Tropical Deserts

These are located on the western margins of the continents between 15°-30° North and South of equator. These regions have hot and dry summers and receive very less rainfall. Vegetation is very scant, with succulent stems, long roots and leaves are reduced to spines.

Tundra

Tundra covers large area in North Canada, North Russia, North Sweden and Finland. The vegetation of tundra is mostly herbaceous, consisting of a mixture of mosses, grasses and forbs, alongwith some dwarf shrubs, trees and lichens. A permanently frozen layer of soil called permafrost restrict growth of plant roots.

AGRICULTURE

- Agriculture is a primary economic activity. It includes both, cultivation of crops and rearing of animals, to provide food and other products.
- Agriculture is practised under different combination of physical and socio-economic conditions which gives rise to different types of agriculture systems.

Types of Agriculture

Shifting Cultivation

- It is also known as slash and burn cultivation as field to be cultivated is cleared by burning the vegetation.
- The cultivated patches are very small and cultivation is done with very primitive tools.
- This type of farming is widely practised by many tribes in the tropics, especially in Africa, South and Central America and South-East Asia.

<i>Shifting Cultivation</i>	<i>Country</i>
Ladang	Malaysia and Java
Tamrai	Thailand
Taungya	Myanmar (Burma)
Caingin	Philippines
Chena	Sri Lanka
Milpa	Mexico
Roca	Brazil
Konuko	Venezuela
Masole	Congo
Tavy	Madagascar
Ray	Vietnam and Laos
Proka	Ghana
Jhuming	India

Intensive Subsistence Agriculture

- This type of agriculture is practised on wide scale in densely populated regions of monsoonal Asia.
- In this type of agriculture the farmer owns a small piece of land, grown crops with the help of his family members and consumes almost the entire farm produce with little surplus to sell in the market.
- This type of agriculture widely practised in Thailand, Myanmar, Vietnam, Cambodia and India.

Plantation Agriculture

- Plantation agriculture is a form of commercial farming where crops are grown for profit. Large land areas are needed for this type of agriculture.
- It was introduced by the Europeans in colonies situated in the tropics. The major products of this type of farming are the tropical crops such as tea, coffer, rubber and oil palm etc has developed in parts of Asia, Africa and Latin America.

Extensive Commercial Grain Farming

- It is commercial form of agriculture where entire operations from ploughing to harvesting are mechanised. Grain crops, especially wheat is grown under extensive farming methods.
- This kind of farming is done by developed countries like USA, Australia and Russia.

Mixed Farming

- In this type of agriculture, equal importance is given to both crop cultivation and animal husbandry.
- It give earning to the farmers.

Nomadic Herding

- It is a primitive subsistence activity, based upon rearing of animals on natural pastures.
- People in this system keep on moving along with their animals in search of grazing areas.
- In Africa, cattle are the most important livestock, while in Sahara and Asiatic deserts, sheep, goats and camels are reared.
- In the mountainous areas of Tibet and Andes, Yak and llamas are reared and in arctic and sub-arctic regions, reindeer is most important animal.

Mediterranean Agriculture

- Specialised commercial farming practised in Mediterranean climatic region. Crops like wheat, barley and vegetables are grown for domestic consumption, while citrus fruits, olives and grapes are grown on commercial scale for export purpose.
- Main regions include countries on either side of Mediterranean in Europe, Southern California, Central Chile, South-Western South Africa and Western Australia.

Market Gardening

- It is concerned with cultivation of high value crops like vegetables, fruits and flowers for urban markets. It is also known as Truck Farming.
- It is well developed in densely populated industrial districts of North-West Europe, North Eastern USA and Mediterranean regions.

Types of Cultures	Description
Apiculture	Raising of bees for commercial purpose
Aquaculture	Rearing of aquatic animals or plants for food
Floriculture	Cultivation of flowers
Horticulture	Cultivation of fruits, flowers and vegetables on commercial scale
Mariculture	Rearing of fish or other marine life for food
Pisciculture	Controlled breeding and rearing of fish
Sericulture	Commercial level production of silk worms
Silviculture	Commercial cultivation of trees
Viticulture	Cultivation of grapevines

Crops	Countries Rankwise			
Food Crops				
Rice/Paddy	1. China	2. India	3. Indonesia	
Wheat	1. China	2. India	3. United States	
Millet	1. India	2. Niger	3. China	
Maize/corn	1. United States	2. China	3. Brazil	
Oat	1. Russia	2. Canada	3. Finland	
Pulses	1. India	2. Pakistan	3. Canada	
Fibre Crops				
Cotton	1. China	2. India	3. United States	
Jute	1. India	2. Bangladesh	3. China	
Plantation Crops				
Rubber	1. Thailand	2. Indonesia	3. Vietnam	4. India
Tobacco	1. China	2. Brazil	3. India	
Coconut	1. Indonesia	2. Philippines	3. India	
Sugarcane	1. Brazil	2. India	3. China	
Tea	1. China	2. India	3. Kenya	
Coffee	1. Brazil	2. Vietnam	3. Indonesia	

LIVESTOCK FARMING/ ANIMAL HUSBANDRY

- It refers to rearing of animals for food and other human uses. The world 'Livestock' applies primarily to cattle, chickens, goats, pigs, horses and sheeps.
- Poultry farmers keep chickens, turkeys, geese and other birds for meat and eggs.

- Top Beef producer of world are USA, Brazil, European Union, China and India.
- India is the world's largest producer of milk. It is the leading exporter of skimmed milk powder. The world's largest exportes of milk and milk products are New Zealand, the European Union, Australia and the United States.



DAIRY FARMING

- It is highly capital intensive and advanced form of farm activity.
- It is practised mainly near urban and industrial centres.
- Denmark and Sweden have witnessed the maximum development of this type of farming.

- Some animals like sheeps, goats and rabbits reared for wool production also. Australia is the leading producer of wool followed by New Zealand and China. Temperate grasslands like Steppes in Russia, Prairies in North America, Pampas in Argentina, Veldt in South Africa and Downs in Southern Australia supports highly intensive livestock industry.

Fisheries

- Fisheries constitute an important economic activity in the coastal regions of the world, especially in the temperate latitudes. Fisheries can be classified into two groups – fresh water fisheries and marine fisheries.
- Around 90% of the fishes are caught from the sea while remaining is obtained from fresh water. The top ten largest fish producing countries in the world are China, Peru, India, Indonesia, USA, Chile, Japan, Thailand, Vietnam and Russia.
- Major exporters of fish and fishery products in the world are China, Norway, Thailand, USA and Denmark and major importers of fish and fishery products are Japan, USA, Spain, France and Italy.

There are four major fishing grounds in the world :

- North-East Atlantic Region** Dogger Bank located in the North Sea is one of the most important fishing ground. Important species of this region are Cod and Herring.
- North-West Atlantic Region** Grand Bank and Georges Bank are important fishing grounds. Important species of this region are Cod, Lobster, Oyster and Herring.
- North-West Pacific Region** Countries involved in fishing in this region are China, Japan, Russia, North and South Korea.
- North-East Pacific Region** This region include coastal regions of Canada and USA. Important species of this region are Halibut, Sardine, Salmon and Tuna.